Theta-Quench Schwinger Lab: Short Report

Hypothesis:

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Stronger theta quenches increase transition signatures and stabilizer magic;
higher-magic snapshots should be harder for NNQS training (scoped claim).
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Model:

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1+1D lattice Schwinger model (open boundaries), staggered fermions,
Gauss-law-eliminated spin Hamiltonian H = H_pm + H_ZZ + H_Z + const.
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Numerics:

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Small-N exact diagonalization + real-time evolution (dense/Krylov),
Pauli-batched stabilizer Renyi magic (M2), GRU/MADE/RBM/independent NNQS fitt
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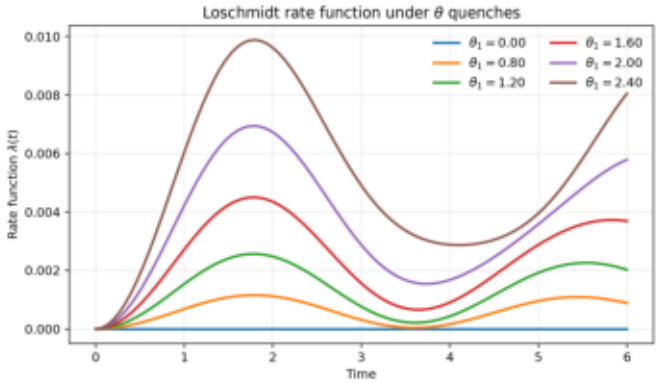
Artifacts:

```
validation: outputs/validation.json
quench summary: outputs/quench_summary.csv
nnqs metrics: outputs/nnqs/nnqs_snapshot_metrics.csv
claim gate: report/publishability_status.md
protocol: report/analysis_protocol.md
```

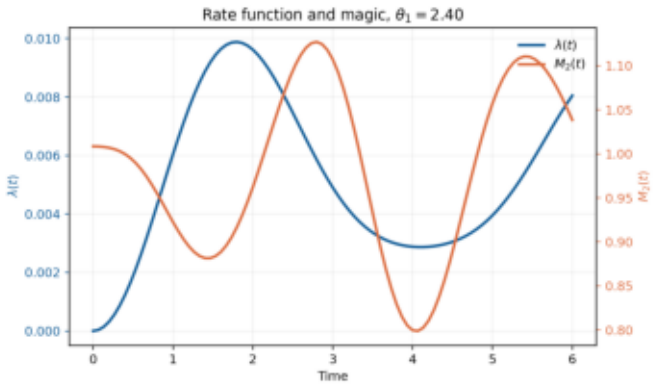
Validation checks:

```
Hermitian H0/H1: True / True
Norm drift: 3.109e-15
Dense-vs-Krylov diff: 9.667e-15
Magic sanity |T>^n: 1.4384103622589088
```

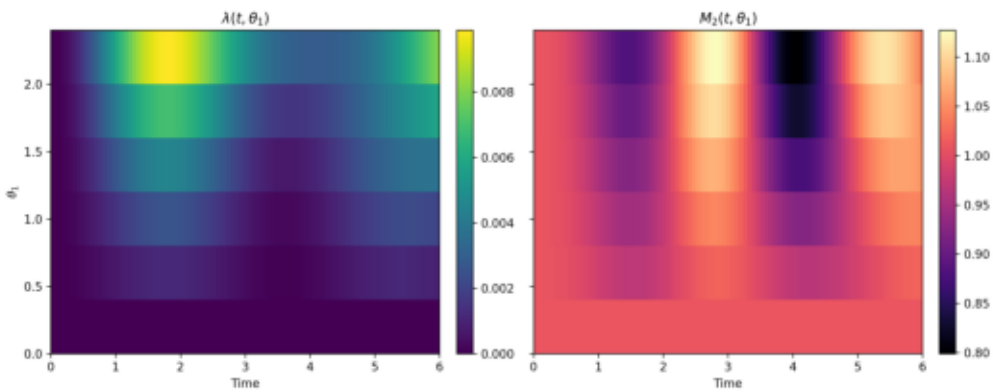
Key Results I: Dynamics + Magic



Loschmidt rate family across quench endpoints.

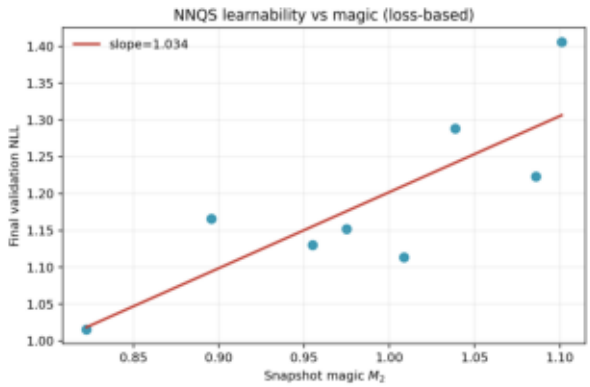


Overlay of transition diagnostic and stabilizer magic M2(t).

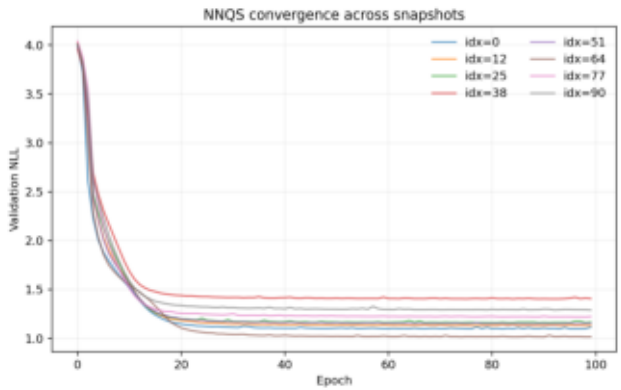


Heatmap view over (time, theta1).

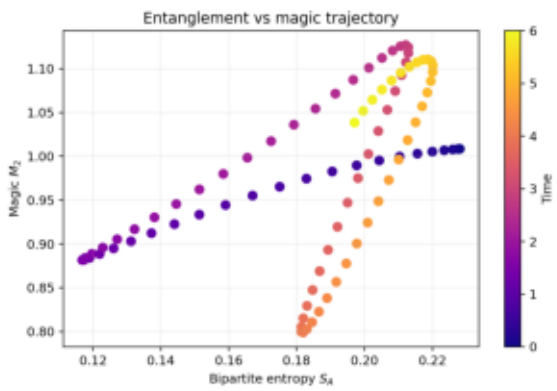
Key Results II: Learnability



NNQS final validation loss vs snapshot magic.

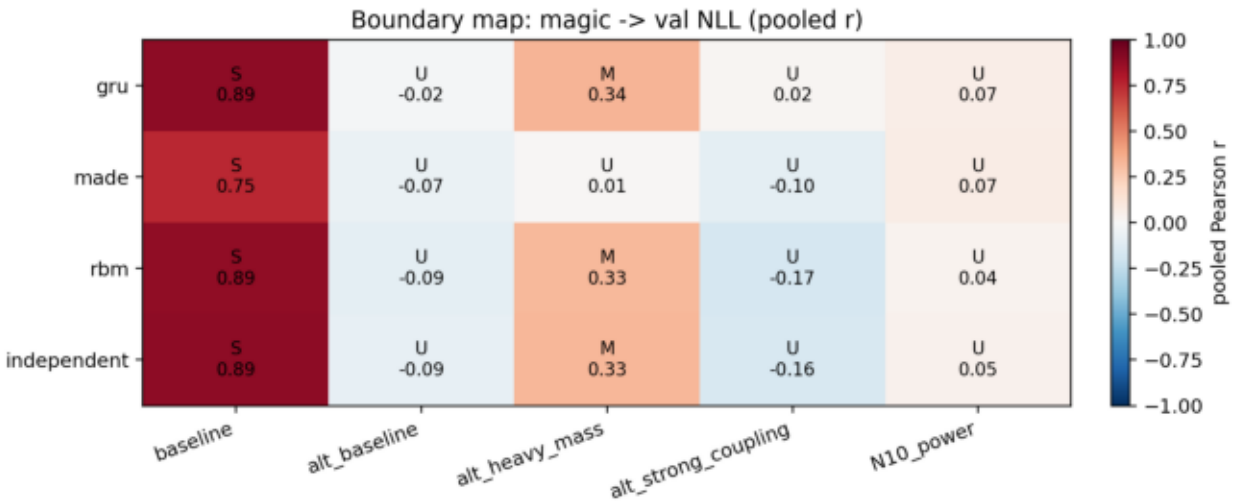


NNQS optimization trajectories over training epochs.



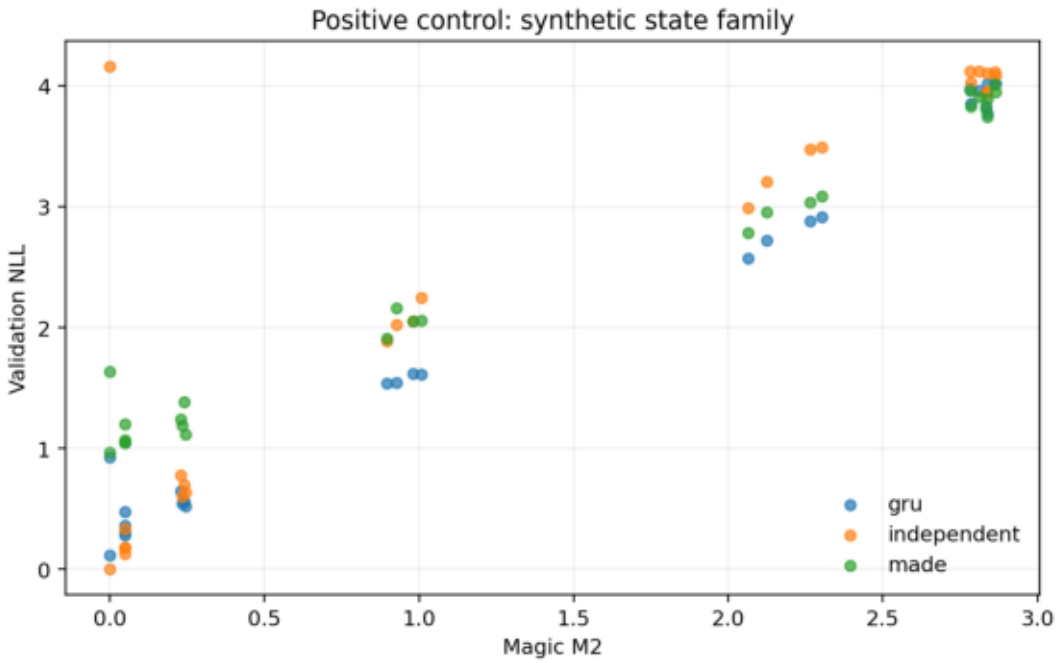
Entanglement vs magic trajectory (complementary diagnostic).

Key Results III: Claim Boundary

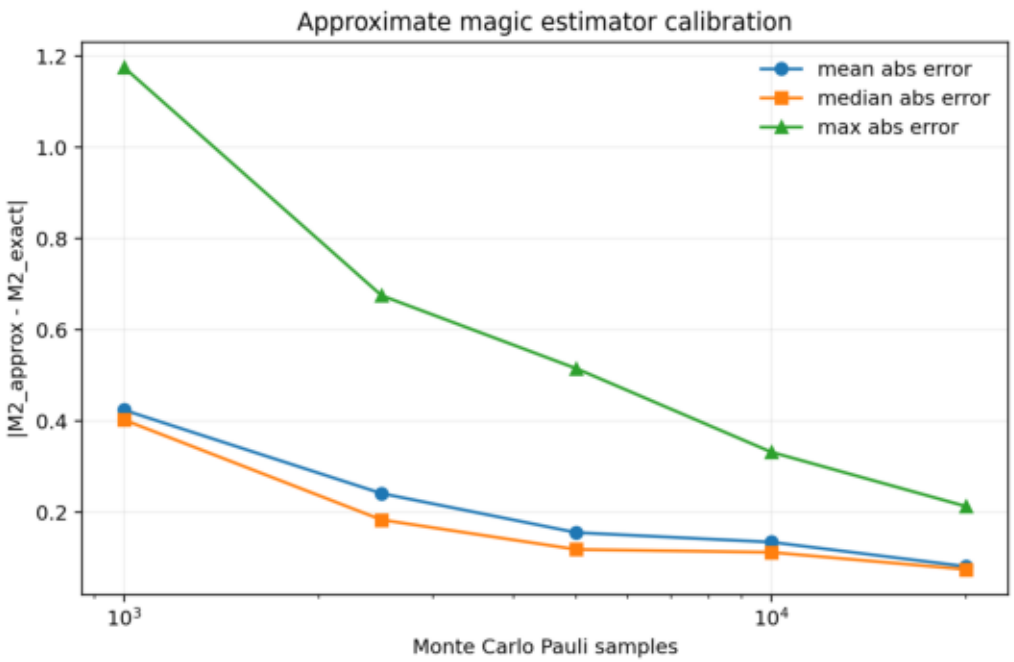


Regime/size boundary map for magic vs validation NLL (conditional, non-universal).

Key Results IV: Controls and Calibration

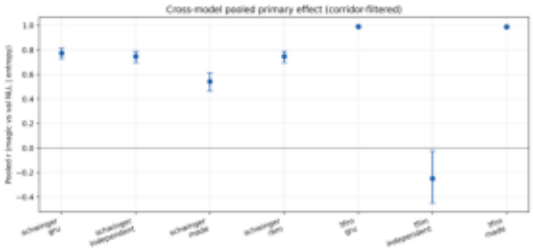


Positive control: synthetic state family confirms increased learnability cost with magic.

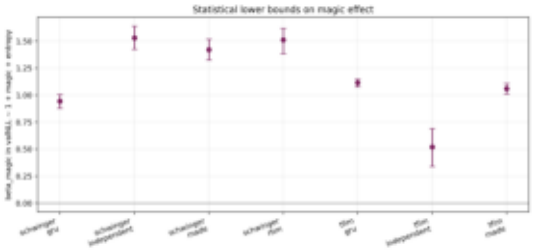


Approximate magic estimator calibration envelope vs exact M2.

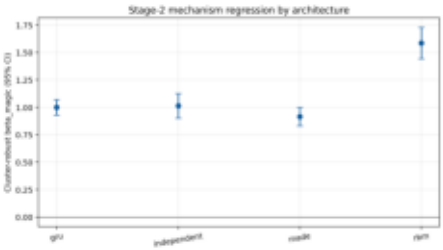
Key Results V: Stage-2 Cross-Model Bounds



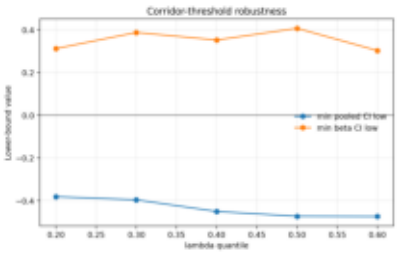
Cross-model pooled primary effects (magic -> validation NLL | entropy).



Bootstrap CI bounds for entropy-controlled magic slope beta_magic.

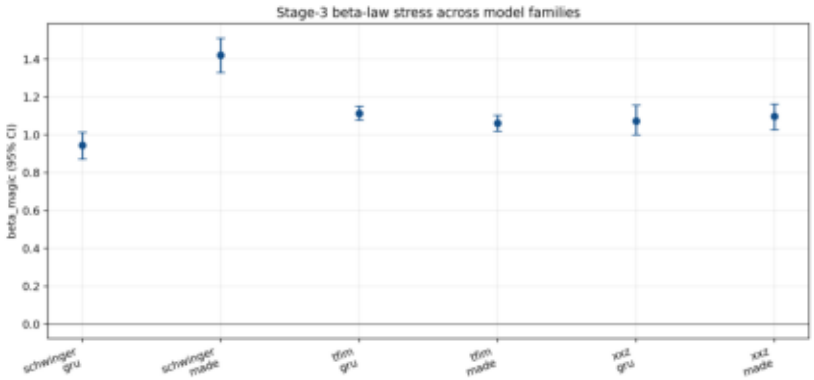


Cluster-robust mechanism regression: positive beta_magic lower bounds by architecture.

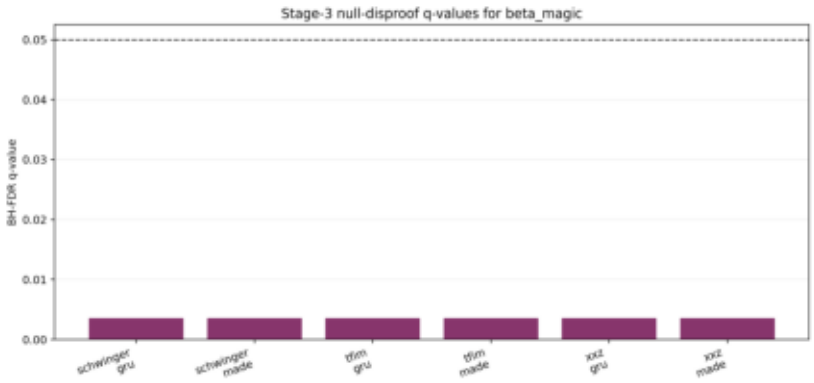


All-architecture robustness: beta-law is stable, correlation-sign law is heterogeneous.

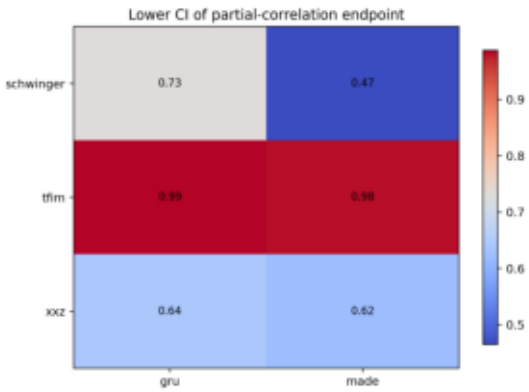
Key Results VI: Stage-3 Null-Disproof Stress



Stage-3 forest plot: beta_magic confidence bounds across Schwinger/TFIM/XXZ cells.

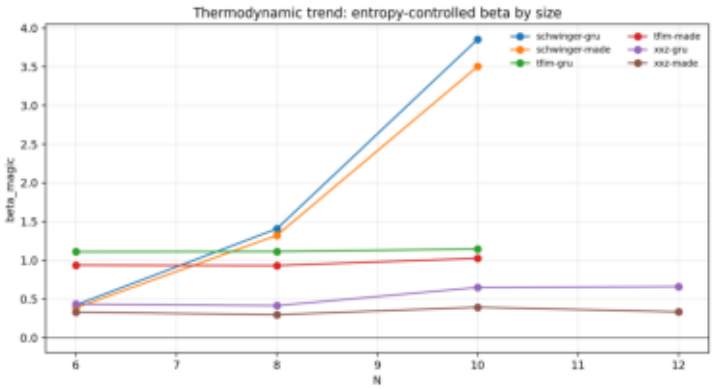


Stage-3 condition-wise permutation null tests with BH-FDR correction.

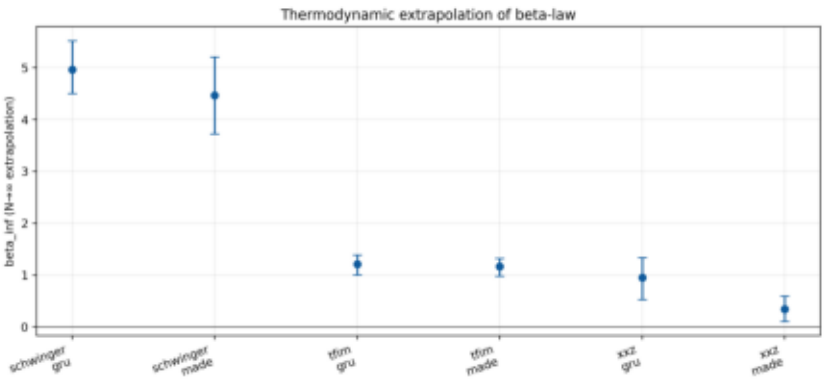


Stage-3 partial-correlation lower-CI heatmap (sign-law heterogeneity map).

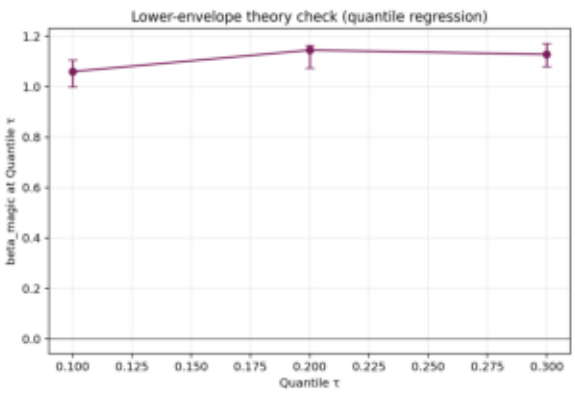
Key Results VII: Stage-4 Thermodynamic + Theory Checks



Finite-size beta-law trend by model family and architecture.

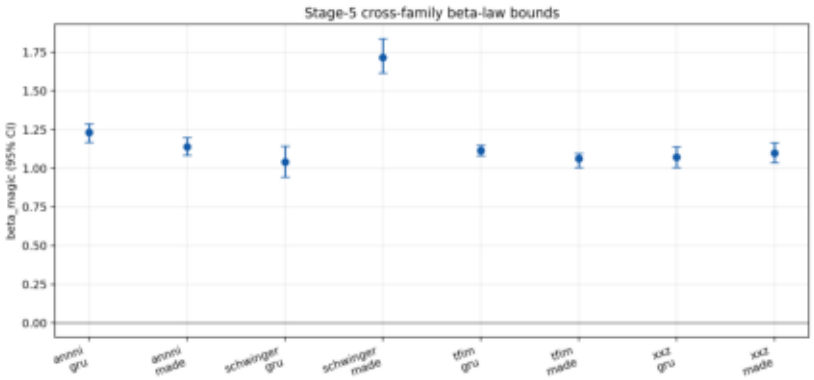


N→∞ extrapolation of beta-law cells with uncertainty.

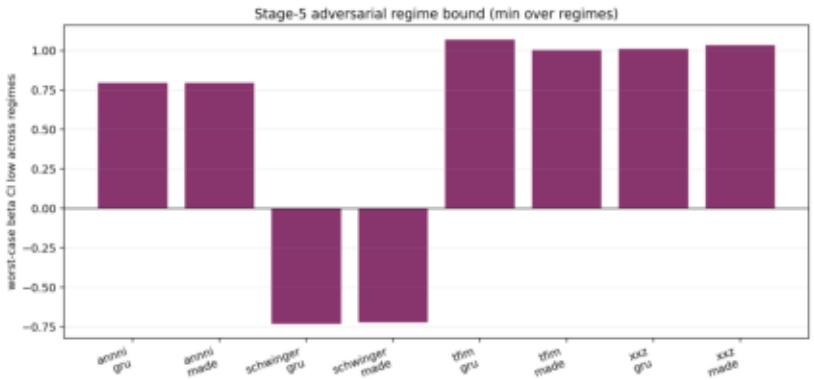


Quantile-regression lower-envelope positivity check for beta_magic.

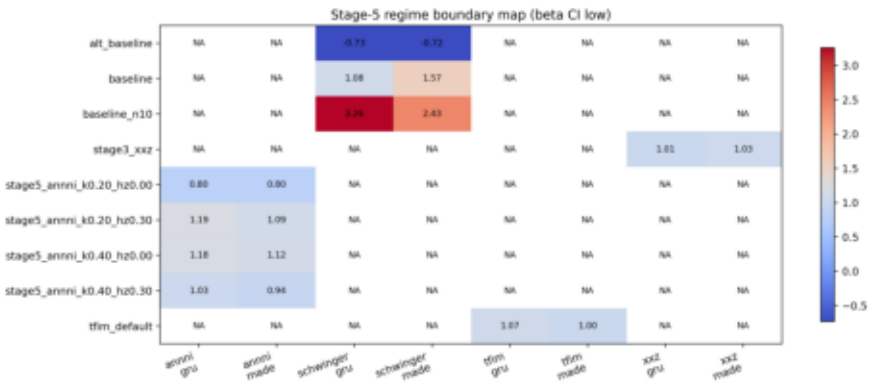
Key Results VIII: Stage-5 External Family + Regime Stress



External-family extension (ANNNI added): cell-level beta-law bounds remain positive.



Adversarial minimax regime bound shows where strict regime-universal positivity fails.



Regime boundary heatmap: explicit support/failure map for beta_magic CI lower bounds.

Roadmap

1) Increase system size via symmetry sectors and sparse block structure.

2) Add sampled/approximate magic estimators for N > 12.

3) Extend NNQS architectures (Transformer/MADE) and measurement bases.

4) Build Trotterized circuit path for hardware-facing experiments.

Caveat:

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Current claims are corridor-scoped empirical laws; not model-independent theo
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